Fundamental Labs II
Assignment from Error Theory
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Assignment

① Least Squares Method:

$y = a + bx$

$\Delta = n\sum x_i^2 - (\sum x_i)^2$

$= 5(220) - (30)^2 = 200$

$a = \frac{1}{\Delta}(\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i)$

$= \frac{1}{\Delta}[(220)(256.6) - (30)(1622)]$

$= \frac{[(220)(256.6) - (30)(1622)]}{200}$

$= 38.96$

$b = \frac{1}{\Delta}(n\sum x_i y_i - \sum x_i \sum y_i)$

$= \frac{1}{200}(5 \cdot 1622 - (30)(256.6))$

$= \frac{1}{200}(= 2.06$

$y = 38.96 + 2.06x$

② $\sigma^2_a = \frac{\sigma^2}{n\sum x_i^2 - (\sum x_i)^2} \sum x_i$

with $\sigma = u^2 = \frac{1}{n-p-1} \sum [y_i - y(x_i)]^2$

$= \frac{1}{5-1-1}[(42-43.00)^2 + (48.4-47.32)^2$
$+ (51.3-51.32)^2 + (56.3-55.44)^2$
$+ (58.6-59.56)^2] = 1.423$

$\sigma_a = \frac{\sigma^2}{n\sum x_i^2 - (\sum x_i)^2} = \frac{1.423^2}{5(220) - 30^2} = 0.007$

$\sigma_b = \frac{n\sigma}{n\sum x_i^2 - (\sum x_i)^2} = \frac{5(1.423)^2}{5(220) - 30^2} = 0.035$

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